Design and Implementation of Support Subsystems for a Homogeneous Electro-Fenton Filter Reactor: Mechanical Supports, PID-Based Heater Control, Pumping System, and PCB Integration

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Abstract— **Textile wastewater in Indonesia carries dyes and complex organic compounds that conventional primary treatment cannot adequately remove. A homogeneous electro-Fenton filter reactor offers an alternative that generates its oxidant in situ and removes the ultraviolet source required by photo-Fenton systems, but the electrochemical process imposes demanding mechanical, thermal, and fluid-handling requirements on the reactor core. This paper presents the design and implementation of four support subsystems for such a reactor. A split-shell three-dimensional printed lid and a custom stand house the electrodes, sensors, and magnetic stirrer while maintaining the required electrode spacing and chemical resistance. An automated pumping system based on seven peristaltic pumps doses reagents and samples and performs flushing, with each channel individually calibrated to its target flow rate. A two-stage controller that combines coarse on-off heating with proportional-integral-derivative regulation holds the reactor within the specified temperature range. Finally, a single printed circuit board consolidates the motor drivers, sensors, and controller for system integration. The subsystems were fabricated and tested, meeting the electrode spacing, flow rate, and temperature targets, while the results also identify the remaining work toward full reactor integration.**

Keywords— electro-Fenton, wastewater treatment, peristaltic pump, PID control, PCB design, 3D printing

I. Introduction

The Indonesian textile industry produces wastewater that contains dyes, suspended solids, and complex organic compounds, which are difficult to remove using primary treatment process [1]. Dyes such as methylene blue, which are commonly found in this wastewater, have detrimental effects on health and the environment, including hindering photosynthesis, causing biomaginfication, and exposing its carcinogenic effects [2]. Regulation of the Minister of Environment and Forestry of the Republic of Indonesia No. 16 of 2019 states that the maximum color concentration of textile wastewater is 200 Pt-Co, meanwhile measurements on actual wastewater from these industries can reach up to 1073.47 Pt-Co [3]. According to this, the degradation rate of color substances in raw textile wastewater must reach at least 81.4%.

Previous developments of filter reactors utilizing zeolite adsorption and photo-Fenton reaction only reached the degradation rate of 50%. Additionally, the reliance on an UV light as a source of energy in the reaction poses challenges in terms of safety and space requirements. TA2552601013 proposes the replacement of the photo-Fenton process by the homogenous electro-Fenton process. This type of Fenton reaction produces H_2O_2 in-situ, while regenerating Fe^{2+} from the residual Fe^{3+} electrochemically, effectively eliminating the need for any UV light source. The proposed design used in order to achieve reliable operation is presented in Fig. 1.

The reactor core is where all of the sensors, actuators, pump pipes, and electrodes are situated in, including temperature sensor, pH sensor, and gas sensor. The actuators are the magnetic stirrer and the heating element of the temperature control system. Since electro-Fenton is an electrochemical process, there is added complexity in due to the addition of electrodes, not to mention the oxygen requirement, further complicating the setup. In order to allow the automation of both the adsorption and electro-Fenton process, there's also a need for an intelligent loading and unloading system for all the required ingredients for the reaction, while keeping the temperature of the core in specific range. This complexity is the reason why a sufficient mechanical accomodation is required. The robustness of the integrated system must be considered as well, with regards to the safety and portability requirements.

This paper presents the design and implementation of four subsystems of the reactor. Firstly, the mechanical aspects of the reactor is implemented as a critical component to the reactor allowing a compact and convenient operation while ensuring robust structural integrity. Secondly, a PID-based heater system is employed to allow the temperature requirement of the specification (25 – 40°C). Thirdly, an automated fluid pumping system consisting of 7 peristatic pumps with specific flow rate to allow the automation of the reactor operation. Finally, a PCB design is presented which contributes to the overall system integration.

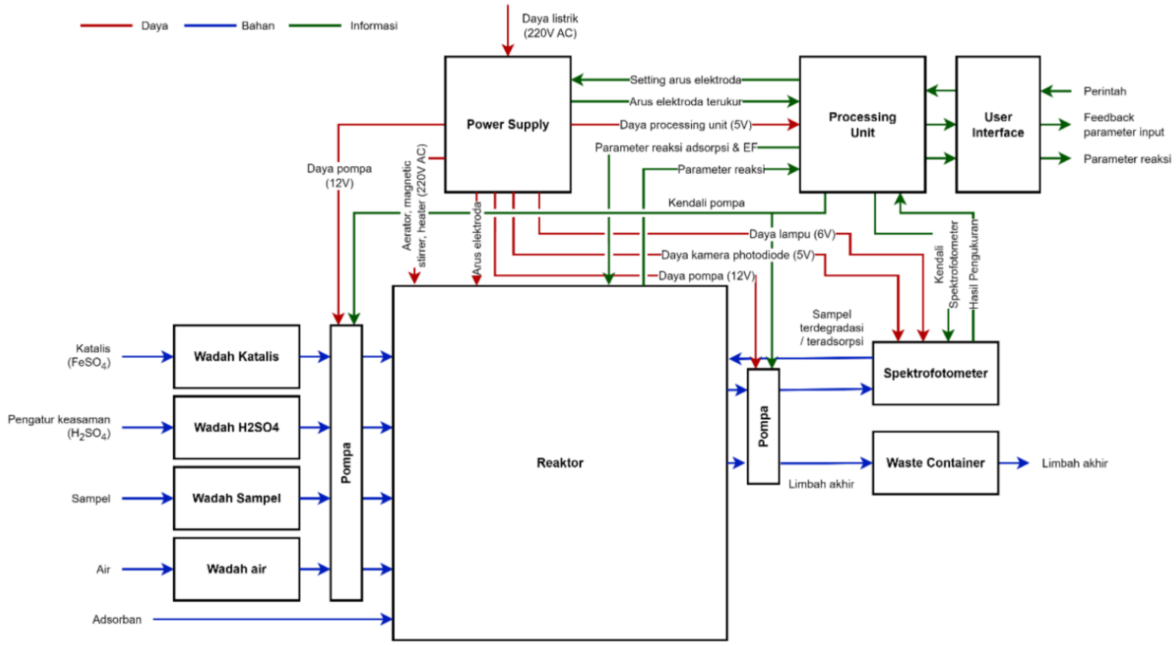

Fig. 1. Adsorption & electro-Fenton main system block diagram

II. Design and Implementation

A. Target Specifications

The specifications and their target values for the proposed subsystems is presented in TABLE I. To achieve an optimal degradation rate, two electrodes with dimensions 150 cm x 100 cm x 0.3 cm needs to be placed 3 cm apart. The design of the reactor must accommodate this while ensuring space availability of all the other components previously described. This includes the aerator which is targeted to provide an airflow of 20 mL is required as the minimum [4], and a maximum of 2000 mL/min is selected in order to provide flexibility. Meanwhile, the pumps is categorized into high-precision pumps that must have a flow rate of 30 mL/min and high-speed pumps that can reach 120 mL/min. This consideration is required to give the precise amounts of ingredients while also optimizing for speed wherever possible. Finally, a variable temperature control needs to be added as well to accommodate reactions in temperatures higher than the surrounding environment, while also keeping it well under the H_2O_2 decomposition temperature [5], [6].

TABLE I. Target Specifications of Proposed Subsystems

Specification	Target
Electrode thickness	150 cm x 100 cm x 0.3 cm
Electrode spacing	3 cm
Aerator air flow	20-2000 mL/min
Reagent and cuvette pump flow rate	30 mL/min
Sample and flushing pump flow rate	120 mL/min
Reactor temperature range	25 – 40° C

B. Components Used

While the popular PLA material can be used to manufacture the reactor supports, the PETG material is ultimately chosen for its superior thermal tolerance. Furthermore, it is also has higher durability, water resistance, and chemical resistance (to acids with a pH of 2–4 containing H_2SO_4 and $FeSO_4$), which are crucial in a product that operates in rigorous conditions.

A peristaltic pump is chosen to move fluid by squeezing a flexible tube with rotating rollers, ensuring no contact between the fluid and the pump components, crucial for preventing cross-contamination between concentrated H_2SO_4 acidic $FeSO_4$ solutions, and dye samples. The Kamoer NKP-DE-S10D is chosen as the high-precision pump, while a generic unbranded peristaltic pump is chosen as the high-speed pump.

For PWM modulation of DC motors, the IRLZ44N MOSFET has a V_{th} of 1–2 V, a drain current up to 47 A, and an economical price of Rp6,000 per unit. The DS18B20 is a 1-Wire digital sensor with an accuracy of ±0.5 °C over the range of -10–85 °C. This sensor was selected due to its low cost (Rp9,450) and its operating range covering the specified 25–40 °C. For the heating element, the built-in heater pad of the SH-2 magnetic stirrer is used along with its included 12 V DC motor.

C. Reactor Mechanical Support

The reactor lid must accommodate six instruments simultaneously: two graphite plate electrodes, a pH sensor, a temperature sensor, a gas sensor, and input/output hoses for the pump. The lid must also be chemically resistant while providing sufficient sealing to contain spills when the magnetic stirrer is in operation.

Key design considerations for the reactor lid are:

- Hole tolerances: 0.5 mm for the holes for the electrodes, gas sensor, pH sensor, and temperature sensor; hose holes have no tolerance because the hoses are flexible.
- Distance between electrodes: maintained at 30 mm according to B200 specifications to accommodate the magnetic stirrer (length ~30 mm) without interference.

The final result of the design iteration process is a split-shell design as shown in **Error! Reference source not found.**. It is split into two parts: (1) the top part which hosts and secures all of the sensors, cables, and hoses, and (2) the bottom part which isolates the content of the reactor as well as holding the electrode in place.

In order to accommodate all of the instruments for the reactor core, the top part forms a 63 mm tall enclosure with 3 mm walls and an upward taper that provides clearance for the electrode alligator clamps. A single opening with a separator routes the hoses, instrument cables, and their protective sleeve, while the separator keeps the fluid lines isolated from the electrical cables and a dedicated channel guides the hoses clear of the electrodes during assembly. A 4 mm fillet on the inner and outer edges removes sharp corners, and the piece attaches to the bottom shell through a cantilever snap-fit.

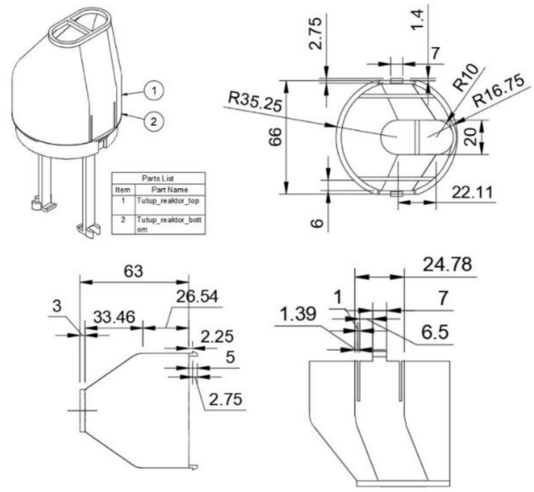

Fig. 2. Assembled reactor core lid and the top piece of the split-shell design

The lower part of the lid seals the reactor and holds two graphite plates 30 mm apart as per the specification. Each slot for the graphite is 3.5 mm thick and has a 70 mm tall holder. This piece also has a built in clip for output hoses and aerator so that it stays at the bottom of the reactor. This allows the reactor to be emptied out, even when the content volume is low.

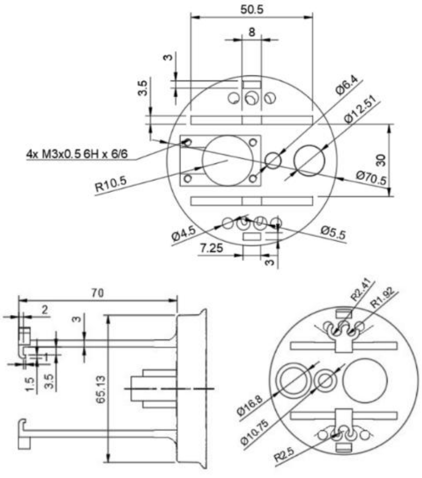

Fig. 3. Reactore core lid bottom piece design

To reuse the magnetic stirrer without the original enclosure, a custom reactor stand is designed to support the reactore core along with the heating pad. As shown in the attached figures, the stand itself is a four-legged frame onto which the motor mount, the heating pad, and the ring holder is installed. The motor mount has its center cleared in order to give room for the magnetic stirrer shaft. Meanwhile, the ring holder keeps the reactor in place despite the torque produced by the stirrer, while also keeping the reactor fully accessible.

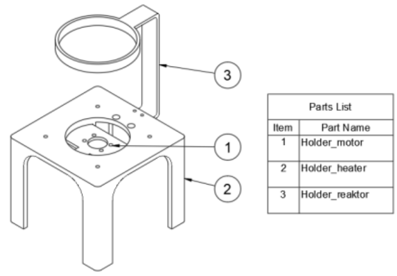

Fig. 4. Assembled reactor stand integrating the motor mount, heater holder, and reactor ring holder

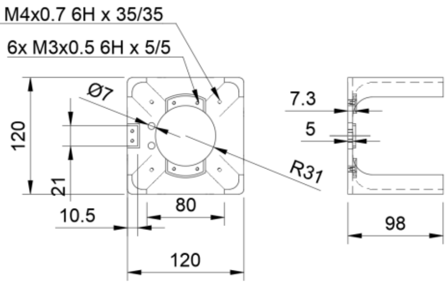

Fig. 5. Heater holder that seats the hot-plate heating element

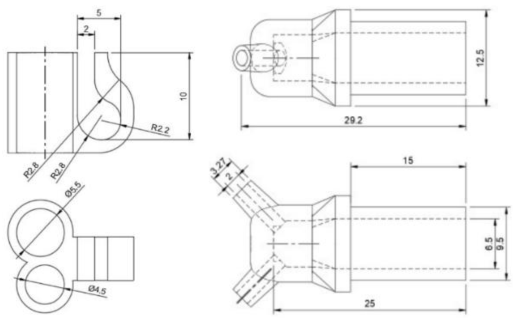

Fig. 6. Hose clip with an integrated tube gripper and the cuvette lid with inlet and outlet ports

Two smaller printed parts completes the fluid handling system. A clip that grips the hose into the external container ensures that it stays attached to the 250 mL beaker, while also ensuring it can pull fluids from the ingredient containers. A cuvette lid is also designed to allows automatic fluid circulation through the spectrophotometer by including two ports for the input and output hoses.

D. Pump System

Fig. 7 shows how every fluids flows through the reactor system and where each pump is positioned in the plumbing. Five input hoses deliver fluids into the reactor, each with their independent pumps: air (aerator), sample, water, H_2SO_4, and $FeSO_4$. The sample and water hoses gets merged using a T-junction first before entering the reactor, while the rest supply the reactor directly. On the output side, one pump moves the content of the reactor into the spectrophotometer for measurement and then loops back. Another pump is dedicated to flushing out the content of the reactor into the waste container.

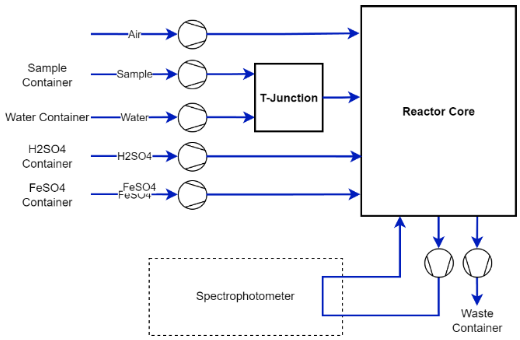

Fig. 7. Reactor pump subsystem fluidic architecture

Each unit pump is controlled using a motor driver as illustrated in Fig. 8. The left part of the figure shows the overall block diagram of the unit, where a motor driver controls the 12 V input into a PWM that matches the control input. On the right is the schematic for the motor driver, which is implemented by the IRLZ44N N-MOSFET. A 100 kΩ gate pull-down holds the transistor off whenever the controller output is unpowered or floating, preventing spurious dosing, while a 1N4007 flyback diode clamps the inductive spike produced when the pump motor is switched off and a 100 nF capacitor suppresses switching noise. This same driver is used for all of the pumps in Fig. 7.

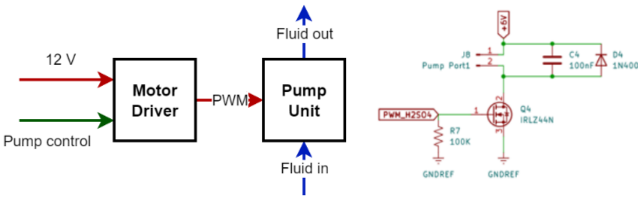

Fig. 8. Electrical driver for a single pump

The pump managemen task is designed around the non-idealities of the chosen peristaltic pumps. Because the pumps, especially the low cost ones usually stalls at PWM values below 170 even at 6 V, each dosing cycle is preceded by a 100 ms long full duty cycle kick. This is intended to built up momentum, especially when there is significant height that needs to be passed by the fluids. In addition to that, since the fluids need time to go through the initially empty hoses, each of them is pre-filled during the system boot so that the dosing volume is not affected. This task then enters a continuous loop where it waits for initialization commands to start to dose the sample, H_2SO_4, and $FeSO_4$ according to the user input. At the same time, it is waiting for the flush command as well, where the system is rinsed with water, including the spectrophotometer's cuvette. In every iteration during the execution phase of the reactor, this subsystem also controls the cuvette and aerator, ensuring it runs continuously. The PWM values used for each pump is adjusted individually because the pumps don't give a consistent PWM-to-flow rate relationship among the different units.

E. Temperature Control

The temperature control architecture is shown in **Error! Reference source not found.** and is implemented as a closed-loop feedback chain. Its component is chosen based on the need to drive an AC heater using a low voltage contoller. A solid-state relay bridges between the logic-level PWM and the 220 AC mains, giving isolation and silent, wire-free switching. The heating element itself is the a heating pad that is coincidentally included with the magnetic stirrer, and the temperature sensor acts as the feedback loop which retrieves the actual temperature of the reactor to be compared with the target setpoint.

As shown in **Error! Reference source not found.**, the temperature control mechanism works as a periodic loop that controls an active-low SSR using PWM signal. In each iteration, the system reads the temperature and only starts the heater when the setpoint is non-zero positive. When it is on, the control is divided into two stages. If the temperature is more than 10 °C under the setpoint, it operates in coarse stage where the heater is fully on. Once the measured temperature reaches that band, the system switches onto the PID stage. This mechanism is chosen because relying only on the PID output would heat up the reacter slowly.

The two stages operates at different frequencies. The coarse stage is updated about every second, while the PID is only updated once every minute. This is because the heater system is identified using data that is sampled once every 60 seconds. The PID output is also clamped to the minimum of 4, since any value lower than that would not turn on the heater.

F. PCB Design

The PCB integrates the motor drivers, sensors, and reactor's actuator into a single board, built using KiCad based on the system's global schematics. The ESP32 and ADS1115 analog-to-digital converter is mounted on pin headers, not directly on the board to optimize replaceability. Each connector is chosen based on the type of signal it is transmitting, either a 2-, 3-, or 4-pin screw terminal for high-current input and output that demands robustness such as the pump and stirrer motors as well as the 12 V, 6 V, and 5 V inputs. Meanwhile, the sensor and display interface uses the 2.54-pitch pin headers that is also compatible with common JST connectors. The board features a ground copper fill and is routed vertically for the top layer and horizontally for the bottom layer for convenience. It measures roughly 134 x 109 mm, complete with M3 screw hols on each corner and silkscreen annotation. This design has also shown to pass the DRC against JLCPCB's fabrication capabilities.

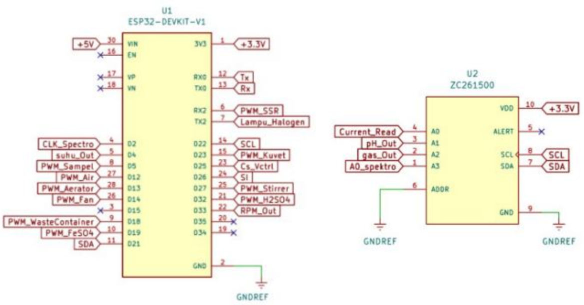

Fig. 9. ESP32 and ADS1115 pinout mapping schematic

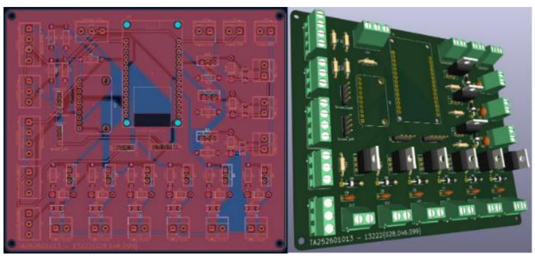

Fig. 10. PCB routing result and the 3D model with components

III. Results and Analysis

A. Reactor Mechanical Design

The split-shell lid and stand for the reactor were printed in PETG and installed onto the reactore. The Elegoo Rapid PETG has been proven to be able to withstand the electro-Fenton solution for 24 hours, and its flexibility allows for snap-fit joints to be implemented. Together, this produces a tight and secure enclosing for the electrodes, cables, and hoses on top of the reactor core. On the custom stand, the magnetic stirrer and heater is able to operate normally while isolating the heated pad from susceptible components. The specified electrode spacing of 30 mm is achieved, now the target submerged area of 6 x 9 x 0.3 cm is only achieved partially. The 0.3 cm thickness is fulfilled, but the 6 cm width is constrained by the beaker's size and electrode availability. Meanwhile, the 9 cm submersion depth is limited by the working volume of 150 mL.

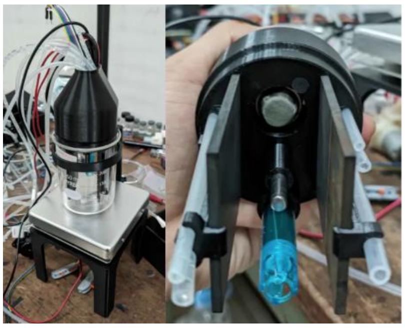

Fig. 11. Assembled electro-Fenton reactor with the split-shell lid mounted on the custom stand

B. Pump Subsystem

Calibration results show that the low-cost generic pump does not have a consistent PWM-to-flow relationship between units, with some of them showing unpredictable behavior as seen in Fig. 13. Because of this, each of them needs to be calibrated using different PWM values. Through several iterations, the PWM values that produces target flow rate is shown in TABLE II.

TABLE II. PWM Calibration Results of Peristaltic Pumps

Pump	Target flow rate	Calibrated PWM value
Sample	120 mL/min	127
Water		129
Waste Container		124
H_2SO_4	30 mL/min	170
$FeSO_4$		187

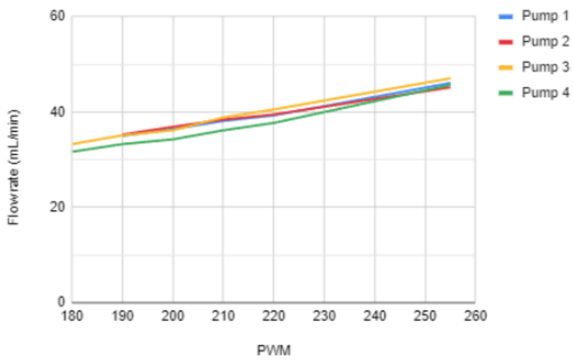

Fig. 12. Precision pump measured flow rate versus PWM

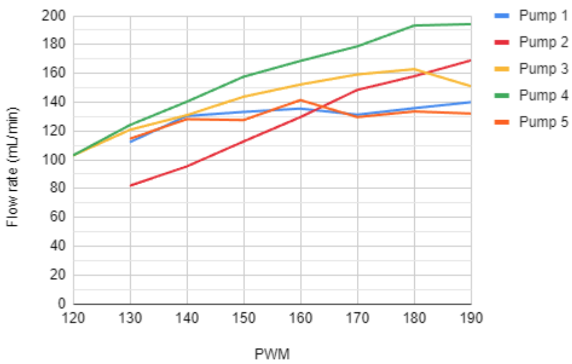

Fig. 13. High-speed pump measured flow rate versus PWM

The H_2SO_4 channel produces satisfying accuracy and precision without much adsjutment using PWM 170, matching the calibration result using a standalone program, while the $FeSO_4$ channel initially doesn't reach the target with the same PWM value despite the identical setup. Troubleshooting effort reveals that the actual flow rate is dependent on the PWM ferequency. It is suspected that the MOSFET gate drive has finite rise and fall time, making the PWM output trapezoidal and reducing the effective duty cycle delivered to the pumps. At 1 kHz, a PWM 170 signal gives out 80% duty cycle, but at 5 kHz it drops to 67-68 %, lowering the terminal voltage and resulting flow rate. Raising the $FeSO_4$ PWM to 187 restored the target flow rate, with rigorous verification still in progress.

Another thing to note is that the T-junction merging the sample and water channels is not optimal for this application because it allowed about 3.32 % of the pumped sample (51.5 mL of 1553.0 mL) to leak into the water container. A Y-junction is expected to guide the flow better towards the correct end of the channel, which is the reactor core. Additionally, the pre-fill duration at boot has been shown to remove the empty-line transport delay and improved dosing repeatability.

C. PID Temperature Control

When driven with full power from the AC mains, the heater successfully raised the reactor temperature to 91.5 °C, which proves that the system has sufficient margin above the specified 25-40 °C. The DS18B20 is accurate enough to match a reference thermometer within 1.5 °C after applying a -1.5 °C offset. The plant transfer function is obtained using MATLAB System Identification tool with a 60 s sampling rate, and tuning with MATLAB SISO tool gives optimal PID parameters of $K_p = 6.977$, $K_i = 6.46 \times 10^{-3}$, and $K_d = 1302.40$. During the first attempt of applying these values, a positive steady-state error is produced at 40 °C, where the controller tries to heat up the reactor despite already reaching the setpoint. Separating the refresh rate of the coarse stage and the PID stage into 1 s and 60 s respectively produces a satisfactory steady state at 40 °C.

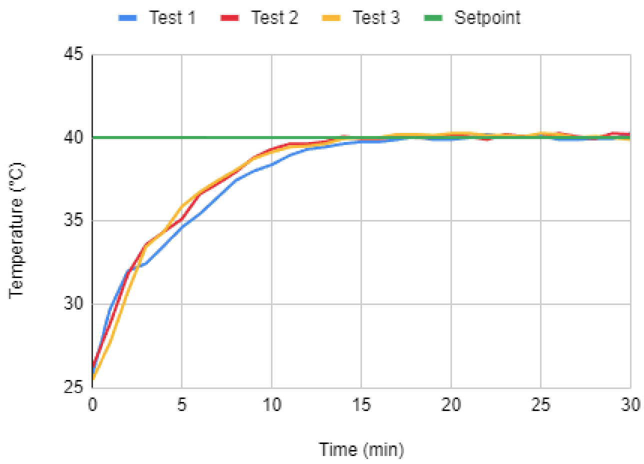

Fig. 14. Closed-loop reactor temperature and heater duty at 40 °C setpoint

D. PCB Integration

The manufactured double-layer PCB has been entirely soldered as seen in **Error! Reference source not found.**. Currently, the JST connector is replaced by standard 2.54 pin header. The PCB has also been show to be able to drive sufficient current to all of the motor pumps simultaneously.

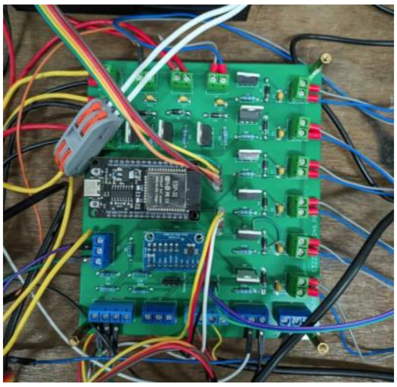

Fig. 15. Assembled reactor control system

E. Summary of Specification Achievements

Specification	Target	Result
Electrode thickness	150 cm x 100 cm x 0.3 cm	Only 0.3 cm thickness met, width and depth limited by working volume
Electrode spacing	3 cm	Met (3 cm)
Aerator air flow	20-2000 mL/min	Adjustable airflow provided; quantitative verification pending
Reagent and cuvette pump flow rate	30 mL/min	Met for H_2SO_4; $FeSO_4$ met after duty adjustment; cuvette pump not yet finalized
Sample and flushing pump flow rate	120 mL/min	Met
Reactor temperature range	25 – 40° C	Regulated within range, with up to 91.5 °C available

IV. Conclusions

This study presented four supportive subsytems for a homogenous electro-Fenton reactor: the mechanical supports, the PID-based heater control, the pumping system, and the integrated PCB. The split-shell PETG lid and stand can support the reactor core with its electrodes and instruments, ensuring a spacing of 3 cm and remain chemically stable in the reaction medium, despite the submerged area constrained by the 150 mL working volume. The pump network automates the fluid logistic and flushing using seven independent pumps with specific PWM per channel, achieving specification after the frequency-dependent caveats has been addressed. The two-stage temperature control adjusts the reactor temperature within 25-40 °C using tuned parameters. Finally, an integrated PCB has been manufactured and installed with components. Future works includes rigorous verification of the pump subsystem, replacing the T-juction with a Y-junction to eliminates the ~3.3 % cross leak, sealing the cuvette lid for watertight spectrophotometer, as well as the full integration of the system.

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